# Chapter 1

# Posterior Inference for Treatment Effects

## 1.1 Overview

The preceding chapters established a structural decomposition of the observed outcome into a prognostic baseline, a causal treatment effect, and a confounding bias term, together with a Bayesian nonparametric prior over the component functions. That development concerns *how the model is fitted*. The present chapter concerns *what is done with the fit*: how the posterior distribution of the component functions is converted into causal conclusions about treatment effects, how the contribution of the real-world data is controlled and measured, and how the heterogeneity of both the treatment effect and the confounding bias is summarised in an interpretable way.

Two estimands are the focus throughout: the conditional average treatment effect (CATE) and the acceleration factor, its exponential. Both are functionals of a single component of the decomposition, the treatment-effect function, and this makes posterior inference for them direct.

The chapter is organised around four tasks. Section 1.2 defines the CATE and the acceleration factor from the fitted decomposition and fixes the target population. Section 1.3 describes posterior summaries of conditional and population-averaged effects, including the treatment of covariate-distribution uncertainty through the Bayesian bootstrap. Section 1.4 introduces the adaptive borrowing architecture: two scalar parameters that govern, respectively, how strongly the prognostic surfaces of the two sources are pooled and how strongly the real-world treatment data are allowed to inform the causal effect. Section 1.5 develops a suite of diagnostics that report how much information was borrowed and where. Finally, Section 1.6 introduces variable importance measures for treatment effect heterogeneity and, exploiting the decomposition, a novel variable importance measure for the confounding function itself.

## 1.2 Causal Estimands: the CATE and the Acceleration Factor

### 1.2.1 The decomposition and its parameterisation

We work with the structural model for the log-survival outcome $Y = \log T$,

$$Y_i \;=\; m_0(X_i, S_i) \;+\; \tau(X_i)\, A_i \;+\; (1 - S_i)\, A_i\, c(X_i) \;+\; \sigma\, \varepsilon_i, \tag{1.1}$$

where $A \in \{0,1\}$ is the treatment indicator, $S \in \{0,1\}$ is the data-source indicator with $S = 1$ for the randomized controlled trial (RCT) and $S = 0$ for the observational study (OS), and $X \in \mathcal{X} \subseteq \mathbb{R}^p$ is a vector of baseline covariates. Here $m_0(\cdot,\cdot)$ is the baseline (control) outcome surface, $\tau(\cdot)$ is the treatment-effect function, and $c(\cdot)$ is the confounding function, which enters only for treated observational units. The residual $\varepsilon$ has a mean-zero, unit-variance distribution with cumulative distribution function $\Phi$, itself assigned a nonparametric prior, and $\sigma > 0$ is a scale parameter.

Two modelling choices, used throughout this chapter, refine (1.1). First, the baseline surface is parameterised as

$$m_0(x,s) \;=\; f(x) \;+\; (1-s)\,g(x), \tag{1.2}$$

so that $f$ is a baseline surface common to both sources — interpretable as the baseline of the RCT population, $s = 1$ — and $g$ is an OS-specific deviation. Second, the confounding function is split into a covariate-homogeneous intercept and a covariate-varying remainder, in scale-separated form,

$$c(x) \;=\; c_0 \;+\; \sigma_c\,\tilde{c}(x), \tag{1.3}$$

where $c_0 \in \mathbb{R}$ is an unshrunk intercept, $\tilde{c}$ carries a standard ensemble prior centred at zero, and $\sigma_c \geq 0$ is a global scale. The intercept $c_0$ absorbs any uniform component of the confounding bias, while $\sigma_c$ controls the magnitude of its variation across covariate values; the rationale for this split is given in Section 1.2.4. The parameters $\sigma_g$ (the leaf scale of $g$) and $\sigma_c$ are the borrowing controls and are the subject of Section 1.4.

### 1.2.2 The conditional average treatment effect and the acceleration factor

The primary conditional estimand is the CATE on the log-time scale,

$$\tau(x) \;=\; \mathbb{E}\big[Y(1) - Y(0) \,|\, X = x\big] \;=\; \mathbb{E}\big[\log T(1) - \log T(0) \,|\, X = x\big]. \tag{1.4}$$

Under the decomposition (1.1), $\tau(x)$ is the treatment-effect function itself: it is a component of the model rather than a derived functional, and it does not depend on the baseline surface, the scale $\sigma$, or the error distribution $\Phi$.

Because the model is of accelerated failure time form, exponentiating the log-scale effect yields the natural multiplicative estimand. We define the acceleration factor

$$\mathrm{AF}(x) \;=\; \exp\{\tau(x)\}. \tag{1.5}$$

The acceleration factor is the factor by which treatment multiplies survival time: $\mathrm{AF}(x) > 1$ indicates that treatment prolongs survival at covariate value $x$, and $\mathrm{AF}(x) < 1$ that it shortens it. It is reported on the time scale on which clinical conclusions are usually framed and, like the CATE, is a function of $\tau$ alone.

**Remark 1.1** (Robustness to the error model)**.** Neither (1.4) nor (1.5) involves the error distribution $\Phi$. The nonparametric prior on $\Phi$ governs the shape of the survival curves and contributes to the efficiency of the fit, but the target estimands are pure functionals of the treatment-effect function. Conclusions about the CATE and the acceleration factor are therefore insensitive to the specification of the error distribution. Should survival-probability contrasts or restricted-mean survival differences be required as secondary summaries, they are recovered from the full model through $\Phi$; they are not pursued in this chapter.

### 1.2.3 Target population and the role of confounding

The causal estimand is the treatment effect in the observational (target) population. By the cross-source exchangeability assumption the treatment-effect function $\tau$ does not depend on the data source, so the CATE and the acceleration factor are common to both populations; what is specific to the target population is the covariate distribution over which conditional effects are averaged, addressed in Section 1.3.

The confounding function does not enter the causal estimands, but it governs the relationship between the causal effect and what is *observed* in the observational study. In the treated observational arm the conditional mean implied by (1.1) is

$$\mathbb{E}[Y \mid X = x, A = 1, S = 0] \;=\; m_0(x, 0) + \tau(x) + c(x),$$

so the observed treated–control contrast in the observational data identifies $\tau(x) + c(x)$, not $\tau(x)$. An analysis of the observational study alone would therefore report a confounded CATE $\tau(x) + c(x)$ and, on the multiplicative scale, a confounded acceleration factor

$$\exp\{\tau(x) + c(x)\} \;=\; \mathrm{AF}(x) \cdot \exp\{c(x)\}.$$

The confounding thus acts *additively* on the CATE scale and *multiplicatively* on the acceleration-factor scale, with $\exp\{c(x)\}$ the factor by which the naive observational acceleration factor is inflated. The role of the decomposition, and of the borrowing architecture of Section 1.4, is to isolate $\tau(x)$ from $c(x)$ so that the reported CATE and acceleration factor are the de-confounded, causal quantities.

### 1.2.4 Interpretation of the confounding function

The confounding function is modelled as a function of the observed covariates $x$, whereas the confounding itself originates in unmeasured covariates $U$ that are by definition absent from $x$. This invites a natural objection: in giving $c$ a prior over functions of $x$ alone, is some assumption being smuggled in about the relationship between $U$ and $X$? The answer is that no such assumption is being made — the dependence of $c$ on $x$ is forced by its definition — but the structure of $U$ does control how the function behaves, and this is worth making explicit.

**The confounding function is a function of $x$ by construction.** As defined in the preceding chapter, the confounding function is a contrast of conditional expectations taken *given* $X = x$; the unmeasured covariates have already been integrated out in forming those expectations. The result is therefore necessarily a function of $x$, and modelling it with an ensemble prior over functions of $x$ is exact rather than approximate. Equivalently, the bias of the observational treated–control contrast, viewed as a function of $x$, is *by definition* equal to $c(x)$: the entire confounding bias present in the conditional mean projects onto a function of the observed covariates, and no component of it escapes into a direction orthogonal to $x$. The framework does not assume the confounding to be small, or to be explained by $x$; it represents it exactly.

**The shape of $c$ is governed by interaction between $U$ and $X$.** What the unmeasured structure controls is not whether $c$ depends on $x$, but how. Consider the prognostic component of the bias, and write the structural prognostic function as $f_0(x, u)$ (the prognostic function of the underlying structural model, distinct from the surface $f$ in (1.2)). Let

$$\Delta(u \mid x) \;=\; p(u \mid x, A{=}1, S{=}0) - p(u \mid x, A{=}0, S{=}0)$$

denote the difference between the treated and control conditional densities of the unmeasured covariates at covariate value $x$; this difference integrates to zero over $u$. The prognostic confounding is then $\int f_0(x,u)\,\Delta(u\,|\,x)\,du$. If the prognostic surface is additive, $f_0(x,u) = a(x) + b(u)$, the $a(x)$ term integrates against $\Delta$ to zero and the expression reduces to $\int b(u)\,\Delta(u\,|\,x)\,du$, which depends on $x$ only through the tilt $\Delta(u\,|\,x)$. If, in addition, that tilt does not depend on $x$, the confounding is a *constant*. The covariate dependence of $c(x)$ therefore arises precisely from interaction between $U$ and $X$ — either through non-additivity of the prognostic surface in $(x,u)$, or through covariate dependence of the treatment-assignment mechanism — and not from mere marginal association between the two. This is the rationale for the intercept–remainder split (1.3): the intercept $c_0$ carries the covariate-homogeneous bias that survives under additive structure, while the shrunk remainder $\sigma_c \tilde{c}(x)$ carries the interaction-driven variation, which is often plausibly small and is therefore an appropriate target for regularisation.

**What the unmeasured covariates still cost.** Although modelling $c$ as a function of $x$ is exact, three consequences of the unmeasured confounding remain and should be borne in mind. First, the ensemble prior and the scale $\sigma_c$ encode a belief that $c$ is regular and ideally small or smoothly varying; whether that belief holds is itself governed by the $U$–$X$ interaction structure above, and a confounding mechanism with complex interaction will be captured poorly, allowing residual bias to leak into $\tau$. Second, the confounding function conflates two sources of bias: an imbalance in the baseline prognostic surface across treatment arms, and — when the treatment effect itself depends on $U$ — an imbalance in effect modification. These two are not separately identified, so $c$ is properly read as the total observable footprint of confounding rather than as prognostic imbalance alone. Third, and most consequentially, the separation of $\tau$ from $c$ rests on the cross-source transportability assumption, which is itself a condition on $U$.

**Remark 1.2** (Transportability is the binding assumption on $U$)**.** Identification of $\tau$ and $c$ requires that the conditional law of the potential outcomes given $X$ be common to the two sources, which in turn requires the conditional distribution of $U$ given $X$ to be comparable across the trial and the observational study. If the two populations differ on an unmeasured axis, so that $U\,|\,X$ is not transportable, then $\tau(x)$ is no longer common to the sources and the confounding function will absorb both genuine within-source confounding and cross-source non-transportability, with no internal means of distinguishing them. This is the assumption the framework cannot diagnose from the observed data, and it is where the relationship of the unmeasured covariates to the data *sources*, rather than to $X$, becomes critical. Sensitivity analysis with respect to this assumption is an essential companion to the analysis.

## 1.3 Posterior Summaries of Treatment Effects

Posterior inference proceeds from $R$ draws of the Markov chain Monte Carlo sampler. Each draw $r = 1, \ldots, R$ supplies a complete set of component functions and scale parameters,

$$\{\, f^{(r)},\ g^{(r)},\ \tau^{(r)},\ c^{(r)},\ \sigma^{(r)},\ \Phi^{(r)} \,\},$$

with $m_0^{(r)}(x,s) = f^{(r)}(x) + (1-s)g^{(r)}(x)$. For the estimands of this chapter only the draws of $\tau$ are needed.

### 1.3.1 Conditional estimands

For a fixed covariate value $x$, the posterior of the CATE is read directly off the treatment-effect draws, $\{\tau^{(r)}(x)\}_{r=1}^{R}$, and the posterior of the acceleration factor is obtained by the monotone transform $\mathrm{AF}^{(r)}(x) = \exp\{\tau^{(r)}(x)\}$. The posterior mean and equal-tailed credible intervals follow from the empirical summaries of these draws; because the transform (1.5) is monotone, a credible interval for the acceleration factor is the exponential of the credible interval for the CATE.

### 1.3.2 Population-averaged estimands

The population estimand is the average of the conditional effect over the covariate distribution of the observational population. The population CATE is

$$\bar{\tau} \;=\; \mathbb{E}\big[\tau(X) \,|\, S=0\big]. \tag{1.6}$$

The corresponding population acceleration factor requires a choice, because exponentiation does not commute with averaging. We define the population acceleration factor as the exponential of the population CATE,

$$\overline{\mathrm{AF}} \;=\; \exp\{\bar{\tau}\} \;=\; \exp\big\{\mathbb{E}[\tau(X) \,|\, S=0]\big\}, \tag{1.7}$$

the geometric mean of the conditional acceleration factors. This is the coherent multiplicative counterpart of the population CATE and is the summary reported below. The arithmetic-mean alternative $\mathbb{E}[\exp\{\tau(X)\} \,|\, S=0]$ is also available from the same draws; it exceeds (1.7) by a Jensen gap and is the appropriate summary only when an average over individuals of the multiplicative effect itself is the quantity of interest.

### 1.3.3 Covariate-distribution uncertainty

A plug-in estimate of (1.6) averages $\tau^{(r)}$ over the observed covariate values $\{x_i : S_i = 0\}$, treating them as fixed. For a model *parameter* this is appropriate, but $\bar{\tau}$ is a *functional of the covariate distribution*, and the observational sample is itself only a finite draw from the target population; conditioning on the empirical covariate distribution understates the posterior uncertainty of $\bar{\tau}$ and of $\overline{\mathrm{AF}}$.

We propagate this additional uncertainty with the Bayesian bootstrap (Rubin, 1981), following its use for causal functionals by Antonelli et al. (2019). A noninformative Dirichlet posterior is placed on the target covariate distribution: at draw $r$, weights

$$w^{(r)} = \big(w_1^{(r)}, \ldots, w_{n_0}^{(r)}\big) \;\sim\; \mathrm{Dirichlet}(1, \ldots, 1)$$

are drawn over the $n_0$ observational units, and the population CATE is formed as the reweighted average

$$\bar{\tau}^{(r)} \;=\; \sum_{i:S_i=0} w_i^{(r)}\, \tau^{(r)}(x_i), \qquad \overline{\mathrm{AF}}^{(r)} \;=\; \exp\{\bar{\tau}^{(r)}\}. \tag{1.8}$$

The resulting posteriors reflect both model uncertainty, through the treatment-effect draws, and covariate-sampling uncertainty, through the Dirichlet weights. The procedure is summarised in Algorithm 1.9.

**Algorithm 1.9: Posterior inference for the population CATE and acceleration factor.**

(1) For each posterior draw $r = 1, \dots, R$:

   (a) evaluate the treatment-effect function $\tau^{(r)}(x_i)$ at every observational unit $i$ with $S_i = 0$;

   (b) draw Dirichlet weights $w^{(r)} \sim \text{Dirichlet}(1, \dots, 1)$ over the $n_0$ observational units;

   (c) form the reweighted population CATE $\bar{\tau}^{(r)}$ and the population acceleration factor $\overline{\text{AF}}^{(r)}$ via (1.8).

(2) Report the posterior means $R^{-1} \sum_r \bar{\tau}^{(r)}$ and $R^{-1} \sum_r \overline{\text{AF}}^{(r)}$, and the empirical quantiles of $\{\bar{\tau}^{(r)}\}_r$ and $\{\overline{\text{AF}}^{(r)}\}_r$ as credible intervals.

## 1.4 Adaptive Borrowing of Real-World Data

The motivation for combining the two data sources is efficiency: although the CATE is identified from the RCT alone, the observational study carries additional information that can sharpen its estimation. The amount of information transferred — the *borrowing* — should not be fixed in advance but determined by the data. This section describes how borrowing is parameterised and shows that a single scalar governs how strongly the observational treatment data inform the causal effect.

### 1.4.1 Two channels of information sharing

The observational study contributes to the fit through two distinct channels, each governed by one of the functions introduced in (1.2) and (1.3).

**Prognostic channel.** Through (1.2), the observational controls inform the shared baseline $f$, while the source-specific deviation $g$ absorbs any systematic difference between the two baseline surfaces. Under cross-source exchangeability, $g$ captures only covariate-shift and measurement-related discrepancies in the prognostic surface and is expected to be small and smooth. The leaf-scale parameter $\sigma_g$ controls its magnitude; assigning it a half-Cauchy prior, $\sigma_g \sim \mathcal{C}^+(0, \alpha_g)$, allows the data to interpolate between $\sigma_g \to 0$, under which the prognostic surfaces of the two sources are fully pooled, and large $\sigma_g$, under which they are estimated separately.

**Confounding channel.** Through (1.3), the treated observational units inform the confounding function $c$. Its variation scale receives a half-Cauchy prior, $\sigma_c \sim \mathcal{C}^+(0, \alpha_c)$, with $\sigma_c = 0$ corresponding to confounding that is uniform across covariates — carried entirely by the intercept $c_0$ — and $c_0 = 0$ additionally to its complete absence.

These two channels are deliberately separated: discrepancy in the baseline prognostic surface and discrepancy due to confounding are conceptually distinct phenomena and are assigned independent priors. The prognostic channel is where the observational study is most useful and least dangerous; the confounding channel is where its contribution must be regulated.

### 1.4.2 Borrowing on the treatment effect is governed by the confounding scale

The central structural fact of the decomposition is that the borrowing of information about the *treatment effect* is controlled entirely by the shrinkage of the confounding function.

**Proposition 1.1** (Confounding scale as the borrowing control)**.** *Consider the conditional mean of the treated observational arm implied by* (1.1)*,*

$$\mathbb{E}[Y \mid X = x, A = 1, S = 0] \;=\; m_0(x, 0) + \tau(x) + c(x).$$

*If $c$ is left unconstrained, the treated observational units identify only the sum $\tau(x) + c(x)$ and contribute no information about $\tau$ separately; the CATE is then identified by the RCT alone. As the prior on $c$ shrinks it towards the zero function, the treated observational units increasingly behave as unconfounded and contribute directly to the estimation of $\tau$. Consequently the scale $\sigma_c$ in* (1.3) *acts as the effective borrowing-strength parameter for the treatment effect: the limit $\sigma_c \to 0$ corresponds to full pooling of the two sources for $\tau$, and the limit $\sigma_c \to \infty$ corresponds to RCT-only inference.*

The justification is an identification argument. Within the observational study the treated–control contrast identifies $\tau(x)+c(x)$ but not its two components separately; separation requires the external information supplied by the RCT. A flexible, unconstrained $c$ therefore absorbs whatever the treated observational data would otherwise contribute to $\tau$, leaving the RCT as the sole source of information about the treatment effect. Shrinking $c$ removes this absorptive capacity in proportion to $\sigma_c$, so that the observational treated arm is progressively pooled with the trial.

Proposition 1.1 has a useful practical reading. Two analyses a practitioner might perform without a formal model — pooling the two datasets and ignoring confounding, or analysing the trial in isolation — are precisely the two endpoints $\sigma_c = 0$ and $\sigma_c \to \infty$. The proposed model does not commit to either: it places a prior over the continuum between them and lets the data select the operating point. It also shows why a likelihood-level downweighting of the observational study, such as a power prior, is not the appropriate instrument here: such a device would simultaneously discount the prognostic channel, where the observational data are most valuable, whereas the relevant control acts only on the confounding channel.

**Remark 1.3** (Effect of the intercept on the borrowing limits)**.** Under the intercept–remainder parameterisation (1.3), the limit $\sigma_c \to 0$ leaves the confounding at the constant $c_0$ rather than at zero. Because $c_0$ is a single scalar, it is identified from the contrast between the RCT and the observational treated arm and is estimated regardless of $\sigma_c$. In this limit the observational treated units therefore still contribute fully to the *covariate-varying structure* of $\tau$ — its heterogeneity — while a single global offset is removed by $c_0$. The statement of Proposition 1.1 thus refines to: $\sigma_c$ governs the borrowing of treatment-effect *heterogeneity*, with the covariate-homogeneous level handled separately by $c_0$. The limit $\sigma_c \to \infty$ is unchanged, and continues to correspond to RCT-only inference for $\tau$.

### 1.4.3 Specification of the borrowing priors

The half-Cauchy scale $\alpha_c$ encodes the analyst's prior stance on confounding. A small $\alpha_c$ expresses skepticism that the observational study is confounded and therefore encourages borrowing; this is the stance of the commensurate-prior tradition for historical-data borrowing (Hobbs et al., 2011). A larger $\alpha_c$ is agnostic and lets the data determine the degree of borrowing, in the spirit of data-fusion estimators. Because the two stances can yield materially different conclusions, both should be implemented and contrasted, and $\alpha_c$ may be calibrated against the range of confounding magnitudes judged plausible — which, on the acceleration-factor scale, can be elicited as a plausible range for the multiplicative bias $\exp\{c(x)\}$. The prognostic scale $\alpha_g$ is less delicate, since exchangeability makes a large prognostic discrepancy implausible, and a weakly informative choice is generally adequate.

**Remark 1.4** (Finer, region-specific borrowing)**.** A single global scale $\sigma_c$ permits only one borrowing strength for the whole covariate space. A finer architecture assigns each tree $t$ in the ensemble for $\tilde{c}$ an inclusion indicator $\gamma_t \sim \text{Bernoulli}(\pi)$, with $\gamma_t = 0$ deactivating the tree and $\pi \sim \text{Beta}(a_\pi, b_\pi)$. The posterior of $\pi$ then reports the fraction of trees required to represent the confounding, and the posterior of $\sum_t \gamma_t$ summarises the complexity of the confounding structure. This extension nests the global-scale specification as the special case $\gamma_t \equiv 1$ and yields additional diagnostics, at the cost of one Bernoulli update per tree.

## 1.5 Diagnostics for Borrowing

A combined analysis of randomized and observational data is only credible if it can report *how much* information was borrowed, *where* in the covariate space the borrowing occurred, and *how sensitive* the conclusions are to that borrowing. Because the borrowing controls of Section 1.4 are explicit model parameters, most diagnostics reduce to interpretable summaries of those parameters and of the functions they scale. We organise the suite by the question each diagnostic answers.

### 1.5.1 How much was borrowed: global scalar diagnostics

**Posterior of the borrowing scales.** The posteriors of $\sigma_c$ and $\sigma_g$ are the most direct summaries. To be interpretable they should be reported in meaningful units. For the confounding channel, summarising the multiplicative bias $\exp\{c(X)\}$ — for example its posterior mean and a central interval over the target population — expresses the confounding on the same acceleration-factor scale as the estimand, in the form "the confounding inflates the observational acceleration factor by a factor between $b_{\text{low}}$ and $b_{\text{high}}$". The split (1.3) additionally allows the homogeneous and heterogeneous parts of the confounding to be reported separately: the intercept $c_0$ as a single uniform bias, and the posterior spread of $\sigma_c\tilde{c}(X)$ as the extent to which that bias varies across patients. The scale $\sigma_g$ is summarised analogously as a prognostic discrepancy between the two sources.

**Effective sample size for the treatment effect.** The practical value of borrowing is the variance reduction it produces. Let $\text{Var}_{\text{full}}\{\tau(x) \mid \mathcal{D}\}$ denote the posterior variance of the CATE under the combined model and $\text{Var}_{\text{rct}}\{\tau(x) \mid \mathcal{D}\}$ the corresponding variance under a refit using the RCT alone. The borrowing is then expressed as an effective number of additional trial patients,

$$\text{ESS}_{\text{add}}(x) \;=\; n_{\text{rct}} \left\{ \frac{\text{Var}_{\text{rct}}\{\tau(x) \mid \mathcal{D}\}}{\text{Var}_{\text{full}}\{\tau(x) \mid \mathcal{D}\}} - 1 \right\}. \tag{1.10}$$

Averaged over the target covariate distribution, (1.10) provides a single headline figure — the borrowing was equivalent to enrolling $\text{ESS}_{\text{add}}$ additional randomized patients — in the idiom established for historical-data borrowing (Neuenschwander et al., 2010; Hupf et al., 2021). Its evaluation requires one RCT-only refit alongside the main analysis. The effective sample size is computed on the CATE scale, on which the variance is naturally defined; the acceleration factor, being a monotone transform, inherits the same conclusion.

### 1.5.2 Where it was borrowed: functional diagnostics

Scalar summaries average over the covariate space and conceal heterogeneity in the borrowing. Two functional diagnostics localise it.

**Borrowing map.** The pointwise posterior of the confounding function — for instance the posterior mean $\mathbb{E}\big[|c(x)|\,|\,\mathcal{D}\big]$, or the exceedance probability $\mathbb{P}\big[|c(x)| > \delta\,|\,\mathcal{D}\big]$ for a clinically meaningful threshold $\delta$ — displayed over the covariate space, identifies the regions in which the model infers confounding. When $p$ is large the map is shown over the leading principal components or conditional on a small set of clinically salient covariates. Presenting the posterior mean alongside the credible-interval width separates regions of inferred confounding from regions of mere uncertainty.

**Local effective sample size.** Evaluating (1.10) as a function of $x$ produces a map of where the observational study contributed and where it did not. Plotting $\mathrm{ESS}_{\mathrm{add}}(x_i)$ against the inferred local bias $|c(x_i)|$ exhibits the adaptive behaviour the framework is designed to deliver: regions of small inferred confounding receive substantial borrowing, regions of large inferred confounding receive little, and the scatter is itself a visual proof of that mechanism.

### 1.5.3 Whether to trust it: robustness checks

**Prior–data conflict.** Overlaying the prior and posterior densities of $\sigma_c$, and of $\sigma_g$, indicates whether the data were informative about the degree of confounding. Near-coincident densities signal that the data carry little evidence either way; a posterior concentrated away from the prior indicates that the data have spoken. Formal prior–data conflict checks (Evans and Moshonov, 2006) can supplement the graphical comparison.

**Tipping-point analysis.** Refitting the model with the borrowing scale fixed at a grid of values, from $\sigma_c = 0$ (full pooling) through the data-driven posterior median to a large value approximating RCT-only inference, and tracing the population CATE or acceleration factor along that grid, reveals the dependence of the conclusion on the degree of borrowing. The data-driven fit should lie between the extremes and the estimand should vary smoothly; an abrupt change, or a reversal of the substantive conclusion in the neighbourhood of the operating point, identifies a fragile result.

**Cross-source posterior predictive checks.** Simulating treated observational outcomes from the posterior predictive distribution and comparing them with the observed treated observational outcomes — for example through an overlay of survival curves with simulation envelopes (Gelman et al., 1996) — provides a model-free check on the confounding correction. Systematic discrepancy confined to the treated arm is the signature of confounding that the model has failed to absorb.

Table 1.1 collects the suite.

Table 1.1: Borrowing diagnostics organised by the question addressed.

| Question | Diagnostic | Form |
|---|---|---|
| How much was borrowed? | Posterior of $\sigma_c, \sigma_g$ | global scalar |
| | Effective sample size, Eq. (1.10) | global scalar |
| Where was it borrowed? | Borrowing map | function of $x$ |
| | Local effective sample size | function of $x$ |
| Can it be trusted? | Prior–data conflict | prior vs. posterior |
| | Tipping-point analysis | estimand vs. $\sigma_c$ |
| | Cross-source predictive check | survival-curve overlay |

## 1.6 Variable Importance for Heterogeneity and Confounding

A flexible estimate of the CATE describes treatment effect heterogeneity but does not, on its own, identify which covariates drive it. Treatment effect variable importance measures (Hines et al., 2022) summarise the contribution of a subset of covariates to the heterogeneity of the CATE through a model-free, nonparametric quantity with an analysis-of-variance interpretation. This section adapts that idea to posterior inference within the present framework and, using the decomposition, extends it to the confounding function.

### 1.6.1 The variance of the treatment effect and the TE-VIM

All expectations and variances in this section are taken over the covariate distribution of the target (observational) population. The measures are computed on the CATE scale, $\tau$, on which the model is additive and the variance decomposition carries its clean analysis-of-variance interpretation; the same covariates govern the heterogeneity of the acceleration factor, which is a monotone transform of $\tau$. The global measure of heterogeneity is the variance of the treatment effect,

$$\mathrm{VTE} \;=\; \mathrm{Var}\{\tau(X)\}, \tag{1.11}$$

which quantifies how much the treatment effect varies across the population (Levy et al., 2021).

**Definition 1.1** (Treatment effect variable importance measure)**.** For a subset $s \subseteq \{1, \ldots, p\}$ of covariate indices, let $X_{-s}$ denote the covariates outside $s$ and define the reduced CATE

$$\tau_{-s}(x_{-s}) \;=\; \mathbb{E}\big[\tau(X) \,|\, X_{-s} = x_{-s}\big].$$

The treatment effect variable importance measure of $s$ is

$$\theta_s \;=\; \mathbb{E}\big[\{\tau(X) - \tau_{-s}(X_{-s})\}^2\big] \;=\; \mathrm{VTE} - \mathrm{Var}\{\tau_{-s}(X_{-s})\}, \tag{1.12}$$

and its standardised form is the proportion $\psi_s = \theta_s/\mathrm{VTE}$.

The two expressions in (1.12) coincide because $\tau_{-s}$ is the $L^2$ projection of $\tau$ onto functions of $X_{-s}$. The measure $\theta_s$ is the increase in mean-squared error incurred when the covariates in $s$ are removed from the conditioning set, equivalently the portion of treatment effect heterogeneity attributable to $s$; the standardised $\psi_s$ reads as the share of VTE explained by $s$, giving the analysis-of-variance interpretation. The construction has been extended to censored survival outcomes by Ziersen et al. (2024), with which the present log-scale CATE is consistent.

### 1.6.2 Posterior inference for variable importance

The variable importance measure of Definition 1.1 is a functional of the CATE, and within the present framework its posterior is obtained directly, without the influence-function machinery and sample splitting required by semiparametric estimators. For each posterior draw $r$:

(1) evaluate the treatment-effect function $\tau^{(r)}(x_i)$ at every target unit;

(2) estimate the reduced CATE $\tau_{-s}^{(r)}(x_{-s}) = \mathbb{E}\left[\tau^{(r)}(X) \mid X_{-s} = x_{-s}\right]$, the conditional expectation of the draw's treatment-effect function given the retained covariates, by a flexible nonparametric regression of $\{\tau^{(r)}(x_i)\}_i$ on $\{x_{-s,i}\}_i$ — equivalently, by marginalising $\tau^{(r)}$ over the conditional covariate distribution of $X_s$ given $X_{-s}$;

(3) form $\theta_s^{(r)} = \sum_i w_i^{(r)} \left\{\tau^{(r)}(x_i) - \tau_{-s}^{(r)}(x_{-s,i})\right\}^2$, with $w^{(r)}$ the Dirichlet weights of (1.8).

The collection $\{\theta_s^{(r)}\}_r$ is the posterior of the variable importance measure, from which the posterior of $\psi_s$ and its credible interval follow. The Dirichlet weights again propagate covariate-distribution uncertainty, so the posterior reflects both model and sampling variability. This posterior-based route is also a substantive advantage: influence-function estimators of variable importance are known to have unreliable calibration at realistic sample sizes, whereas the posterior delivers uncertainty quantification intrinsically.

**Remark 1.5** (The reduction is a conditional mean, not a restricted projection)**.** The reduced function in Definition 1.1 is the conditional expectation $\mathbb{E}[\tau(X) \mid X_{-s}]$, equivalently the $L^2$-optimal predictor of $\tau(X)$ among *all* functions of $X_{-s}$. Step (2) must therefore estimate it with a flexible learner. Restricting the reduction to a linear — or otherwise low-complexity — class would change the estimand from "the heterogeneity explained by $X_{-s}$" into "the heterogeneity explained by a function of $X_{-s}$ within that restricted class", understating importance and potentially reordering the covariate ranking, the more so because the treatment-effect function is itself deliberately nonlinear. The conditional-mean reduction is the convention of Hines et al. (2022) and of the multivariate-treatment importance measures of Shin et al. (2024). A deliberate projection of the CATE onto a linear or sparse summary is a legitimate but *distinct* object — the posterior projection of Woody et al. (2020), or the best partially linear projection of Ziersen et al. (2024) — which answers a different question and is not a substitute for the conditional-mean reduction used here.

### 1.6.3 Strategies for estimating the variable importance measures

The variable importance measures of Definitions 1.1 and 1.2 are population functionals, and several estimation strategies for them appear in the literature. They differ along three axes: how the CATE is estimated, how the reduced conditional mean is obtained, and how the importance functional is assembled and debiased. We summarise the main options and situate the posterior plug-in of Section 1.6.2 among them; the same options apply to the confounding measure with $c$ in place of $\tau$.

**One-step efficient-influence-curve estimator.** The semiparametric route of Hines et al. (2022) constructs the augmented inverse-probability-weighted pseudo-outcome $\varphi(Z)$, which behaves in expectation like the unobserved individual treatment effect, and estimates

$$\widehat{\theta}_s = n^{-1} \sum_{i=1}^{n} \Big[ \{\widehat{\varphi}(z_i) - \widehat{\tau}_s(x_i)\}^2 - \{\widehat{\varphi}(z_i) - \widehat{\tau}(x_i)\}^2 \Big].$$

This is a one-step (estimating-equation) estimator built from the efficient influence curve of $\theta_s$: it is Neyman-orthogonal, supplies a closed-form influence-curve variance, and is paired with cross-fitting to avoid empirical-process (Donsker) conditions. It is not a plug-in estimator and may return negative values. The CATE $\widehat{\tau}$ may be supplied by a T-learner or, for robustness, by the doubly robust DR-learner, and the reduced CATE $\widehat{\tau}_s$ by regressing either $\widehat{\tau}$ or $\widehat{\varphi}$ on $X_{-s}$.

**Plug-in and targeted (TMLE) estimators.** A direct plug-in $\widehat{\theta}_s = \widehat{\mathrm{Var}}\{\widehat{\tau}(X)\} - \widehat{\mathrm{Var}}\{\widehat{\tau}_s(X)\}$ respects the parameter space but inherits the regularisation bias of the nuisance estimators. The debiased plug-in is the targeted maximum likelihood estimator, in which the nuisance fits are updated so that the plug-in is also efficient: Levy et al. (2021) give such an estimator for the VTE and Li et al. (2023) for the TE-VIM. Targeting $\theta_s$ is more delicate than targeting the VTE, because the update must move the full and reduced regressions compatibly.

**Bayesian posterior plug-in.** When a full posterior over the outcome and treatment-effect functions is available, as in the present framework, the importance measure is a deterministic functional of each posterior draw, and its posterior is obtained by evaluating that functional draw by draw — the route of Shin et al. (2024) and the one adopted in Section 1.6.2. Three features make it the natural choice here. First, uncertainty quantification comes directly from the posterior, with no influence curve and no cross-fitting. Second, the reduced and full CATE are automatically *compatible* within a draw: because $\tau_{-s}^{(r)}$ is the conditional mean of the same drawn function $\tau^{(r)}$, the identity $\mathbb{E}\{\tau\} = \mathbb{E}\{\tau_{-s}\}$ holds exactly at every draw — a property that Hines et al. (2022) must engineer through their choice of reduced regression. Third, the regularisation is supplied by the model's own priors; in particular, the shrinkage on the heterogeneity components (Section 1.4) pulls the importance measures toward zero when heterogeneity or confounding is absent. The validity of the posterior plug-in rests on the outcome model being well-calibrated; Shin et al. (2024) establish that posterior contraction for the outcome surface transfers to the importance measures.

**Separate-sample estimation for inference at the null.** At the zero-importance boundary $\theta_s = 0$ the influence curve of the one-step estimator degenerates and asymptotic normality fails (Remark 1.7). Hines et al. (2022) restore valid inference by estimating $\mathrm{Var}\{\tau(X)\}$ and $\mathrm{Var}\{\tau_s(X)\}$ with efficient estimators on *independent* subsamples and differencing them, at the cost of efficiency. The posterior analogue is to avoid testing $\theta_s = 0$ directly and instead compare importances through the posterior of a *difference*, $\psi_{\{j\}} - \psi_{\{k\}}$, which is well behaved away from the joint boundary; this is the testing device of Shin et al. (2024) and the one we use for the effect-modifier-versus-confounder comparisons of Section 1.6.5.

**Computational devices.** Two accelerations are worth noting. Cross-fitting (sample splitting) is required for the one-step estimator and optional but stabilising for the others. For the posterior plug-in, the bottleneck is the per-draw evaluation of the reduced conditional mean over all units, which amounts to an $n \times n$ evaluation of the treatment-effect function at each draw.

The blocking scheme of Shin et al. (2024) partitions the sample into $K$ blocks, computes the importance measure within each block, and averages across blocks; it is near-identical to the full computation in their simulations and is substantially faster. For a data-fusion analysis with a large observational registry, the blocking scheme is the practical default.

**Remark 1.6** (Choosing the covariate subsets)**.** Distinct from how $\theta_s$ is estimated is the choice of which subsets $s$ to evaluate. Hines et al. (2022) identify four modes: leave-one-out ($s$ a single covariate), keep-one-in ($s$ all but one covariate), Shapley aggregation over all subsets, and covariate grouping guided by domain knowledge. When covariates are correlated, single-covariate (leave-one-out) importances are diluted — two correlated effect modifiers each receive a small measure — so keep-one-in or covariate grouping is preferable in that case.

### 1.6.4 A variable importance measure for confounding

The decomposition isolates a second heterogeneity function, the confounding function $c$, and the construction of Definition 1.1 applies to it verbatim. Let $\mathrm{VCF} = \mathrm{Var}\{c(X)\}$ be the total variation of the confounding bias across the target population, and for a subset $s$ define the reduced confounding function $c_{-s}(x_{-s}) = \mathbb{E}[c(X) \mid X_{-s} = x_{-s}]$. Under the parameterisation (1.3) the homogeneous intercept $c_0$ cancels from every variance and centred contrast below, so the confounding variable importance measure speaks only to the covariate-varying part of the confounding — which is appropriate, since a uniform bias is by construction not attributable to any covariate.

**Definition 1.2** (Confounding variable importance measure)**.** The confounding variable importance measure of a covariate subset $s$ is

$$\theta_s^c \;=\; \mathbb{E}\big[\{c(X) - c_{-s}(X_{-s})\}^2\big] \;=\; \mathrm{VCF} - \mathrm{Var}\{c_{-s}(X_{-s})\}, \tag{1.13}$$

with standardised form $\psi_s^c = \theta_s^c/\mathrm{VCF}$.

The measure $\theta_s^c$ quantifies how much of the confounding bias in the observational study is attributable to the covariates in $s$. It answers a question that has no counterpart in the standard variable importance literature — which observed covariates drive the unmeasured confounding — and is definable only because the decomposition exposes $c$ as an estimable function. Its posterior is computed exactly as in the previous subsection, with $c^{(r)}$ in place of $\tau^{(r)}$.

### 1.6.5 Separating effect modifiers from confounding drivers

Computing the single-covariate importances $\psi_{\{j\}}$ and $\psi_{\{j\}}^c$ for each covariate $j$ and plotting them against one another classifies the covariates into four regimes: pure effect modifiers, important for the treatment effect but not for the bias; pure confounding drivers, important for the bias but not for the effect; covariates important for both; and covariates important for neither. This separation is possible only because the decomposition carries $\tau$ and $c$ as distinct functions, and it directly serves the applied reader: it distinguishes covariates that genuinely modify the treatment effect from covariates that merely reflect how patients were selected into treatment in the observational study, a distinction that an analysis of the observational data alone cannot draw.

**Remark 1.7** (The null case)**.** When a covariate subset has no role in the heterogeneity of interest, the corresponding importance measure equals zero, which lies on the boundary of the parameter space. Inference is delicate at this boundary for any estimator. The measures are therefore best reported as a ranking accompanied by posterior credible intervals, with values near zero treated as a boundary case rather than as the basis for a formal test of no effect.

### 1.6.6 Variable importance as a borrowing diagnostic

The variable importance measures also feed back into the diagnostic suite of Section 1.5. Computing $\psi_{\{j\}}$ for the treatment effect under the RCT-only refit and under the combined fit shows whether borrowing sharpens the identification of effect modifiers: tighter posteriors under the combined fit indicate a genuine gain, while a marked change in the ranking of effect modifiers is a warning sign, signalling either informative borrowing or contamination of the trial signal by the observational data and warranting closer inspection through the tipping-point analysis.

## 1.7 Summary

This chapter has set out how the fitted decomposition is turned into inference about treatment effects. The two estimands of interest, the conditional average treatment effect and the acceleration factor, are functionals of the treatment-effect function alone and are insensitive to the error model; their posteriors are read directly off the treatment-effect draws and summarised with the Bayesian bootstrap to account for covariate-distribution uncertainty. The contribution of the observational data is governed by two explicit scale parameters, of which the confounding scale $\sigma_c$ is shown to be the effective borrowing control for the treatment effect, with the confounding acting multiplicatively on the acceleration-factor scale. A suite of diagnostics reports how much was borrowed, where, and how robustly; and variable importance measures summarise the heterogeneity of both the treatment effect and the confounding function, the latter being a measure made possible by the decomposition. Together these elements make the borrowing transparent and auditable, which is a precondition for the use of real-world data in confirmatory treatment effect estimation.

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